ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	High resistance draws almost no current.
2	**(c)** 6 ohm	R2 = rho 2L / 4A = R / 2.
3	**(b)** V against I	Ohm's law.
4	**(c)** A is true but R is false.	A is true; R is false: copper has low resistivity.
5	**(b)** 9 R	L is tripled, A is one third.
6	**(d)** 5 ohm	R = V / I.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Greater collisions give more resistance.
8	**(b)** R / n^2	Each part R/n; parallel gives R/n^2.
9	**(d)** 1.5 A	R = 2 + 2 = 4 ohm.
10	**(a)** Both A and R are true and R is the correct explanation of A.	Parallel arrangement features.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	I = 4.8 x 10^-3 A (1); n = 4.8 x 10^-3 / 1.6 x 10^-19 = 3 x 10^16 (1).	1+1
12	R = 220^2 / 800 = 60.5 ohm (1); H = P t = 800 x 300 = 2.4 x 10^5 J (1).	1+1
13	R = V^2 / P (1); Ratio R100 : R25 = 1 : 4 (1).	1+1
14	Two halves of 2 ohm each in parallel (1); R = 1 ohm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Atoms vibrate more, increasing collisions with electrons (1); Resistance increases (1); Current decreases (I = V / R) (1).	1+1+1
16	P = 300 W (1); I = 300 / 220 = 1.36 A (1); R = 220 / 1.36 = 161.3 ohm (1).	1+1+1
17	Ammeter in series should not add to the resistance (1); Voltmeter in parallel should not draw current (1); Else the readings are altered (1).	1+1+1
	OR: Area ratio = 1 : 4 (1); R1 : R2 = 4 : 1 (1); Parallel: I1 : I2 = 1 : 4 (1).	1+1+1
18	Voltmeter has a very high resistance (1); Total resistance becomes very high (1); Almost no current flows and the bulb will not glow (1).	1+1+1
	OR: R = 4 / 0.2 = 20 ohm (1); I = 10 / 20 = 0.5 A (1); P = 5 W (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Parallel part = 48 / 16 = 3 ohm; total = 9 ohm (1); (b) I = 12 / 9 = 1.33 A (1); (c) Voltage across parallel = 1.33 x 3 = 4 V (1); Currents: 4/12 = 0.33 A and 4/4 = 1 A (1); (d) P = 12 x 1.33 = 16 W (1).	1+1+1+1+1
	OR: 20 ohm: P = 14400 / 20 = 720 W (1); 30 ohm: P = 480 W (1); Series 50 ohm: 288 W (1); Parallel 12 ohm: 1200 W (1); Parallel gives the highest power (1).	1+1+1+1+1
20	New R = 9 x 10 = 90 ohm (1); Each third = 30 ohm (1); 1/R = 3 / 30 (1); R = 10 ohm (1); Resistance comes back to the original (1).	1+1+1+1+1
	OR: R60 = 806.7 ohm; R40 = 1210 ohm (1); Total = 2016.7 ohm; I = 0.109 A (1); V60 = 88 V; V40 = 132 V (1); P60 = 9.6 W (1); P40 = 14.4 W (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 27 W	P = V^2 (1/2 + 1/4) = 36 x 0.75.
2	**(b)** 4 ohm	Each branch 8 ohm; two in parallel gives 4 ohm.
3	**(a)** Both A and R are true and R is the correct explanation of A.	The fuse protects appliances.
4	**(d)** 3 ohm and 6 ohm	3 + 6 = 9 and 18 / 9 = 2.
5	**(b)** 2 Q	R halves; H = V^2 t / R doubles.
6	**(d)** A is false but R is true.	A is false; R is true: parallel resistance is less than each.
7	**(a)** 60 W	Same current, larger R dissipates more power.
8	**(b)** 144 W	Series: V^2/(2R) = 36; parallel: V^2 (2/R) = 4 times.
9	**(c)** A is true but R is false.	A is true; R is false: nichrome has high resistivity and high melting point.
10	**(c)** 0.64 A	(140 W) / 220.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	P is proportional to V^2 (1); P = 100 x (1/4) = 25 W (1).	1+1
12	P = 2.2 kW (1); E = 4.4 kWh, cost = Rs 26.4 (1).	1+1
13	Current divides inversely to resistance (1); I = 4 x 12 / 16 = 3 A (1).	1+1
14	Cold filament has low resistance so a large current flows (1); Sudden heating weakens it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	I = 12 / 6 = 2 A (1); P1 = 4 W, P2 = 8 W, P3 = 12 W (1); Total = 24 W = V I (1).	1+1+1
16	Joule is a very small unit (1); 1 kWh = 3.6 x 10^6 J (1); kWh is convenient (1).	1+1+1
17	Current is lower for the same power (1); Heat I^2 R t in the lines is reduced (1); Less power is wasted (1).	1+1+1
	OR: Stretched resistance = 4R (1); Each half = 2R (1); Parallel: R (1).	1+1+1
18	Resistance decreases as length falls (1); P = V^2 / R increases (1); The coil gets hotter (1).	1+1+1
	OR: V = 2 x 6 = 12 V (1); I in 12 ohm = 1 A (1); Total = 3 A (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Motor: 6 kWh (1); Iron: 0.75 kWh (1); Bulbs: 5 x 0.06 x 6 = 1.8 kWh (1); Daily 8.55 kWh; month 256.5 kWh (1); Bill = 256.5 x 6.5 = Rs 1667.25 (1).	1+1+1+1+1
	OR: Fuse wire melts when the current exceeds a limit (1); It is connected in series (1); Current of the geyser = 2200 / 220 = 10 A (1); A fuse of rating a little above 10 A (e.g. 15 A) is chosen (1); A 5 A fuse would melt (1).	1+1+1+1+1
20	R1 = 2 / 0.4 = 5 ohm (1); R2 = 6 / 1.2 = 5 ohm (1); R is the same, so ohmic (1); P = V I = 6 x 1.2 (1); P = 7.2 W (1).	1+1+1+1+1
	OR: V / I = 10 ohm each time (1); So Ohm's law holds (1); P = 3 x 0.3 = 0.9 W (1); rho = R A / L = 10 x 10^-6 / 5 (1); rho = 2 x 10^-6 ohm m (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Electricity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 800 J	H = I^2 R t; 16 / 4 x 1 / 2 x 400.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Currents are added algebraically (Kirchhoff's rule is not required), so current is a scalar.
3	**(c)** 275 W	P is proportional to V^2.
4	**(d)** 1 kWh	0.25 x 4.
5	**(a)** Both A and R are true and R is the correct explanation of A.	The field moves with a high speed along the wire.
6	**(b)** 5 : 3	R is proportional to 1/P.
7	**(c)** Reducing the temperature	Lower temperature decreases R.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R = V^2 / P.
9	**(a)** four times	P = I^2 R.
10	**(b)** 660 W	220 x 3.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat produced depends on I^2 R t (1); The fuse melts when the current exceeds the limit (1).	1+1
12	R1 + R2 = 25 and R1 R2 = 100 (1); R1 = 5 ohm and R2 = 20 ohm (1).	1+1
13	R = 806.7 ohm (1); I = 110 / 806.7 = 0.136 A (1).	1+1
14	H = I^2 R t (1); H becomes 8 times (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	0.5 (4 + r) = 0.25 (8 + r) (1); 2 + 0.5 r = 2 + 0.25 r (1); r = 0, emf = 2 V (1).	1+1+1
16	Each side 3 ohm (1); One side 3 ohm in parallel with the other three sides 9 ohm (1); R = 27 / 12 = 2.25 ohm (1).	1+1+1
17	Same current flows (1); Nichrome has higher resistance (1); H = I^2 R t is greater for nichrome (1).	1+1+1
	OR: I = 220 / 55 = 4 A (1); P = 880 W (1); E = 0.88 x 5 = 4.4 kWh, cost Rs 22 (1).	1+1+1
18	Total resistance doubles (1); Current halves (1); Each bulb glows dimmer (1).	1+1+1
	OR: Q = 2 x 30 = 60 C (1); V = 20 V (1); E = V Q = 1200 J (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	H = V Q = V I t = I^2 R t (1); P = H / t = 8.4 x 10^5 / 600 = 1400 W (1); I = P / V = 6.36 A (1); R = V / I (1); R = 34.6 ohm (1).	1+1+1+1+1
	OR: Series: R = 20 ohm, I = 1 A (1); Series total power = 20 W (1); Parallel: R = 5 ohm, I = 4 A (1); Parallel total power = 80 W; each bulb 40 W (1); Series: failure of one stops both; parallel: the other continues (1).	1+1+1+1+1
20	(a) All in series: 3 + 6 + 9 = 18 ohm (1); (b) All in parallel: 1/R = 1/3 + 1/6 + 1/9 = 11/18 (1); R = 1.64 ohm (1); (c) 3 ohm and 6 ohm in parallel give 2 ohm (1); Adding 9 ohm in series gives 9 + 2 = 11 ohm (1).	1+1+1+1+1
	OR: At 220 V: I = 22000 / 220 = 100 A (1); Loss = I^2 R = 10000 x 2 = 20000 W (1); At 2200 V: I = 10 A (1); Loss = 100 x 2 = 200 W (1); High-voltage transmission reduces the loss 100 times (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.